

FMFM: A Python Package for Automatically Computing First Motion-based Focal Mechanism Solutions – v1.1 – Dec. 23, 2025

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1. Licence



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2. Introduction

The focal mechanisms of small-to-moderate earthquakes provide vital constraints for the analysis of complex fault structures and regional stress fields. Due to their short rupture durations, these events allow for a robust double-couple point source approximation (Lay and Wallace, 1995). Traditional determination methods, such as the HASH algorithm, rely on P-wave first-motion polarities and S/P amplitude ratios (Hardebeck and Shearer, 2002, 2003). However, in the context of high-density seismic arrays, traditional manual analysis is not only prohibitively inefficient but also susceptible to subjective bias. While deep-learning approaches have recently introduced automation, they frequently struggle with generalization issues when applied to tectonic regions that differ significantly from their original training datasets.

To overcome these challenges, FMFM (First Motion-based Focal Mechanism) offers an automatic, rule-based framework designed for the end-to-end processing of large-scale seismic data. By transitioning directly from raw waveforms to focal mechanism solutions, FMFM integrates the POSE algorithm (Wang et al., 2025) for probabilistic polarity identification, automated S/P ratio calculation and correction, and the HASH inversion module into a single streaming architecture. This integrated workflow eliminates the need for complex, multi-step data exchanges between disparate tools, providing a robust and efficient solution for generating high-resolution focal mechanism catalogs.

This manual is intended to help researchers run HASH on their own data sets. The source code may be obtained at: <https://github.com/longtanwang/FMFM>. The methodology is discussed in more detail in our publications.

3. Installation

FMFM is a software suite designed for focal mechanism determination.

- **Python Environment:** Users must first set up a suitable Python environment. This can be easily accomplished by installing **Anaconda** and the required dependencies.
- **HASH Integration:** Additionally, users need to download the **HASH_1.2** source code (<https://www.usgs.gov/node/279393>), extract it into the `./src` directory, and compile it using the **Makefile** provided in our software package.

4. Getting Started: A Practical Guide

4.1 Data Preparation

FMFM requires **continuous waveform data**, **earthquake catalogs**, and **phase arrival times** as input.

- **Test Examples:** We have provided pre-configured test examples in the `./example_projects` directory.
- **New Projects:** When initiating a new project, please ensure the data is formatted according to these examples.
- **Parameter Configuration:** *
General parameters are set in `./example_projects/config.py`.
Parameters specifically for the HASH focal mechanism inversion are defined in `./example_projects/input/example.inp`.

4.2 Step-by-Step Execution

Follow these steps in order to process your data:

1. **Determine P-wave first-motion polarity:** `python 1.run_POSE.py`
2. **Calculate S/P amplitude ratios:** `python 2.calc_SP_Ratio.py`
3. **Prepare for focal mechanism inversion:** `python 3.prep_HASH.py`
4. **Execute focal mechanism inversion:** `python 4.run_HASH.py`

4.3 Interpreting Results and Quality Control

The output files are stored in the following locations:

Polarity Results: `./example_projects/output/polarity/example_polarity.dat`

Amplitude Ratios: `./example_projects/output/focal_mechanisms/HASH_io/example.amp`
Amplitude Ratio

Corrections: `./example_projects/output/focal_mechanisms/HASH_io/example.statcor`

Final Focal Mechanism

Solutions: `./example_projects/output/focal_mechanisms/example_fms.csv` (Summary table)

`./example_projects/output/focal_mechanisms/example_raw.out` (Raw HASH output)

5. Citation

When using this software, please cite:

L. Wang, H. Meng, Y. Zhou, Y. Hou, W. Pei, and S. Zhou. (2025). FMFM: A Python Package for Automatically Computing First Motion-based Focal Mechanism Solutions, *Seismol. Res. Lett.* under review

Hardebeck, J. L., & Shearer, P. M. (2002). A new method for determining first-motion focal mechanisms. *Bulletin of the Seismological Society of America*, 92(6), 2264-2276.

Hardebeck, J. L., & Shearer, P. M. (2003). Using S/P Amplitude Ratios to Constrain the Focal Mechanisms of Small Earthquakes. *Bulletin of the Seismological Society of America*, 93(6), 2434–2444.

Related publications:

L. Wang, Y. Zhou, H. Meng, W. Pei, and S. Zhou (2025). P-wave First-motion Polarity Determination Using Order Statistics and Entropy Theory (POSE) with Applications to Southeastern Tibetan Plateau. ESS Open Archive.

Pei, W., J. Zhuang, and S. Zhou (2025). Stochastic determination of arrival time and initial polarity of seismic waveform. *Earth, Planets and Space* 77(1), 36.325